From Local Curvature to the Weil Functional:

An Explicit Construction

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Abstract

Building on the curvature decomposition established in [1], we construct an explicit family of admissible test functions $g^*_{\sigma,\varepsilon} \in S_{ad}$ for the Weil explicit formula. For this family we derive a prime-side identity of the form

$$W(g^*_{\sigma,\varepsilon} * \check{g}^*_{\sigma,\varepsilon}) = Z(g^*_{\sigma,\varepsilon}) - H_{\text{local}}(\sigma,\kappa) + O(\varepsilon),$$

where $Z(g^*_{\sigma,\varepsilon})$ denotes the corresponding zero sum. We further introduce renormalized prime weights $c_p^{\text{ren}} > 0$, prove convergence of the associated series

$$D = \sum_p \left(c_p^{\text{ren}}\right)^2 \approx 9.471,$$

and show that the renormalized prime-side scale is logarithmic in κ. This isolates the remaining open problem as a single spectral inequality relating $Z(g^*_{1/2,\varepsilon})$ to $H_{\text{local}}(1/2,\kappa)$. We also record the appearance of the Mertens constant as the natural scale of the renormalized curvature remainder.

1 Introduction and relation to Paper 1

In [1] we studied the curvature field

$$H_\xi(\sigma,t) := \frac{d^2}{d\sigma^2} \log|\xi(\sigma+it)|^2 = H_{\text{local}}(\sigma,\kappa) + H_{\text{dual}}(\sigma,t,\kappa),$$

where $H_{\text{dual}} := H_\xi - H_{\text{local}}$ is the spectral remainder encoding the zero contributions (sign convention as in [1], equation (5)). where the truncated local curvature

$$H_{\text{local}}(\sigma,\kappa) = \psi_1(\sigma) + \sum_{p\le\kappa} V_p(\sigma)$$

contains the archimedean term and the finite-place contributions. The prime-side pieces satisfy $V_p(\sigma) \ge 0$, and on the critical line one has $H_{\text{local}}(1/2,\kappa) \asymp (\log\kappa)^2$.

The central open question of [1, §7] asks whether this local positivity and critical divergence can be packaged into a global Weil-type positivity statement. The present note addresses a concrete part of that question by constructing an explicit family of admissible test functions

$g^*_{\sigma,\varepsilon} \in S_{ad}$ and identifying the corresponding prime-side term in the Weil functional with the local curvature studied in Paper 1.

The remaining open step is then isolated as a single spectral inequality; see Section 6.

2 Construction of the admissible test function

2.1 The raw family

Recall from [1, Eq. (3)] that

$$V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2}.$$

Define

$$c_{p,\sigma} := V_p(\sigma)^{1/2} \cdot \frac{p^{1/4}}{(\log p)^{1/2}},$$

and let

$$\varphi_\varepsilon(x) = \frac{1}{\varepsilon\sqrt{2\pi}} e^{-x^2/(2\varepsilon^2)}$$

be the standard Gaussian bump. For fixed κ set

$$\widehat{f}_{\sigma,\varepsilon}(x) = \sum_{p\leq\kappa} c_{p,\sigma}\, \varphi_\varepsilon(x - \log p).$$

The admissible test function is obtained by antisymmetrization:

$$\widehat{g}^*_{\sigma,\varepsilon}(x) := \widehat{f}_{\sigma,\varepsilon}(x) - \widehat{f}_{\sigma,\varepsilon}(-x). \tag{1}$$

Lemma ($g^*_{\sigma,\varepsilon} \in S_{ad}$)**.** *For every $\sigma > 0$ and $\varepsilon > 0$, the function $g^*_{\sigma,\varepsilon}$ belongs to the admissible class S_{ad} used in Lagarias' formulation of Weil positivity.*

Proof. The Gaussian bumps are Schwartz, and antisymmetrization preserves the Schwartz class. Hence $g^*_{\sigma,\varepsilon}$ and $\widehat{g}^*_{\sigma,\varepsilon}$ are rapidly decreasing. Moreover, $\widehat{g}^*_{\sigma,\varepsilon}(0) = 0$ by antisymmetry, and likewise $g^*_{\sigma,\varepsilon}(0) = 0$. These are the required vanishing conditions at the origin. Finally, the Gaussian decay ensures absolute convergence of the prime-side and zero-side expressions appearing in the Weil functional for this family. □

3 Prime-side identity for the Weil functional

Proposition (Prime-side identity)**.** *For the family $g^*_{\sigma,\varepsilon}$ constructed above, one has*

$$W(g^*_{\sigma,\varepsilon} * \check{g}^*_{\sigma,\varepsilon}) = Z(g^*_{\sigma,\varepsilon}) - H_{\mathrm{local}}(\sigma,\kappa) + O(\varepsilon), \tag{2}$$

where $Z(g^*_{\sigma,\varepsilon}) = \sum_\rho |\widehat{g}^*_{\sigma,\varepsilon}(\rho)|^2$ runs over the nontrivial zeros of $\zeta(s)$.

Proof sketch. Apply the Weil explicit formula in the Lagarias framework. The constant term vanishes because $\widehat{g}^*_{\sigma,\varepsilon}(0) = 0$. By construction of the coefficients $c_{p,\sigma}$, the prime-side quadratic form converges to $H_{\text{local}}(\sigma, \kappa)$ as $\varepsilon \to 0$. The error term $O(\varepsilon)$ arises from the Gaussian smoothing of the prime-power contributions and is not made uniform in κ in the present note. □

Remark. *Equation (2) is the key bridge between the curvature field of Paper 1 and the Weil positivity framework. The archimedean term $\hat{g}^*(0)^2 c_0$ vanishes since $\hat{g}^*(0) = 0$ by condition (S2), so the archimedean contribution to W is zero in the present normalization.*

4 Renormalized weights and convergence

4.1 Positivity of the renormalized coefficients

At $\sigma = 1/2$ define

$$f_p := V_p(1/2) - \frac{4(\log p)^2}{p}.$$

A direct calculation gives

$$f_p = \frac{4(\log p)^2(2p-1)}{p(p-1)^2} > 0 \qquad (p \geq 2). \tag{3}$$

Lemma (Renormalized coefficients are positive)**.** *For every prime $p \geq 2$, the quantity f_p in (3) is strictly positive.*

Proof. The numerator $2p - 1 > 0$ and the denominator $p(p-1)^2 > 0$ for every prime $p \geq 2$. □

The renormalized coefficients are simply

$$c_p^{\text{ren}} := f_p^{1/2} > 0, \tag{4}$$

so that $[c_p^{\text{ren}}]^2 = f_p$. Numerical values for small primes:

p	$V_p(1/2)$	$4(\log p)^2/p$	$f_p = [c_p^{\text{ren}}]^2$
2	3.844	0.961	2.883
3	3.621	1.609	2.012
5	3.238	2.072	1.166
7	2.945	2.164	0.781
11	2.530	2.091	0.439

4.2 Convergence of the renormalized energy

Lemma (Convergence)**.** *The series*

$$D := \sum_p [c_p^{\text{ren}}]^2 = \sum_p \frac{4(\log p)^2(2p-1)}{p(p-1)^2} \tag{5}$$

converges absolutely. Numerically $D \approx 9.471$ *(primes below* 10^4*; tail* < 0.009*).*

Proof. For large p, $[c_p^{\mathrm{ren}}]^2 \sim 8(\log p)^2/p^2$. Since $\sum_n (\log n)^2/n^2 < \infty$, convergence follows by comparison. $\square$

5 The renormalized scale and the Mertens constant

By the prime number theorem,

$$H_{\mathrm{local}}(1/2, \kappa) \sim 2(\log \kappa)^2 \qquad (\kappa \to \infty),$$

so the renormalized remainder

$$H_{\mathrm{local}}^{\mathrm{ren}}(1/2, \kappa) := H_{\mathrm{local}}(1/2, \kappa) - 2(\log \kappa)^2$$

is of lower order than $(\log \kappa)^2$. Numerically, $H_{\mathrm{local}}^{\mathrm{ren}}$ exhibits a sawtooth pattern: it jumps upward at each new prime $p \leq \kappa$ by $f_p > 0$, and decreases continuously between primes as $2(\log \kappa)^2$ grows. This is the direct analogue of the prime-counting function $\pi(x)$, which increases by 1 at each prime and is constant between consecutive primes. The Mertens constant γ_M governs the average drift of this sawtooth via the asymptotic

$$\sum_{p \leq \kappa} \frac{\log p}{p} = \log \kappa + O(1),$$

connecting the renormalized scale to classical prime-sum estimates.

Remark. *The Mertens constant*

$$\gamma_M = \gamma + \sum_p \Big[\log \tfrac{p}{p-1} - \tfrac{1}{p}\Big] \approx 0.2615,$$

where $\gamma \approx 0.5772$ *is the Euler–Mascheroni constant, appears in prime-sum estimates that control the average behavior of* $H_{\mathrm{local}}^{\mathrm{ren}}$*. The same constant* γ *appears in the first Li coefficient* $\lambda_1 = 1 + \gamma/2 - \log 2 - (\log \pi)/2 \approx 0.0231$*, reflecting the universal role of* γ *in the transition from discrete prime sums to continuous logarithmic integrals.*

Remark. *The renormalized construction clarifies the scale of the prime-side contribution, but does not by itself resolve the spectral inequality* (6)*.*

6 The remaining spectral inequality

For the family $g^*_{1/2,\varepsilon}$, Weil positivity reduces to

$$Z(g^*_{1/2,\varepsilon}) \;\geq\; H_{\mathrm{local}}(1/2, \kappa). \tag{6}$$

Open question. Is there an unconditional lower bound

$$Z(g^*_{1/2,\kappa}) = \sum_\rho \left|\widehat{g}^*_{1/2,\kappa}(\rho)\right|^2 \;\geq\; C \cdot (\log \kappa)^2$$

for some constant $C > 0$, where the sum runs over nontrivial zeros of ζ?

Formally, treating the zeros as independent via Selberg's orthogonality principle [8] yields $Z(g^*) \sim D \log \kappa \log\log \kappa$ with $D \approx 9.471$. More precisely: the diagonal contribution

$$\sum_p \left[c_p^{\mathrm{ren}}\right]^2 \cdot I(\varepsilon) \;\sim\; D \cdot \log \kappa$$

is unconditional (it follows from the prime number theorem alone); the off-diagonal control of the cross-terms $\sum_\gamma \cos(\gamma \log(p/q))$ for $p \neq q$ remains open. An unconditional proof of even $Z \geq C(\log \kappa)^2$ is presently unavailable.

Remark (The constant $D/2\pi$)**.** *The asymptotic $Z(g^*) \sim D \log \kappa \log\log \kappa$ yields a natural bridge constant*

$$\frac{D}{2\pi} \approx \frac{9.471}{6.283} \approx 1.507,$$

which measures the ratio of the prime-side energy D to the archimedean normalization 2π of the zero density $N(T) \sim T \log T/(2\pi)$. This constant is of a different nature from the pair-correlation statistics studied in the GUE conjecture (Montgomery 1973): it depends only on the total density of zeros via the classical formula for $N(T)$, which is unconditional and requires neither GUE nor RH as input. Whether $D/2\pi$ admits a representation in terms of classical constants is an open question.

Remark. *By Lagarias [3],* RH $\iff$ $W(f * \check{f}) \geq 0$ *for all $f \in S_{ad}$. A positive answer to the open question above would establish $W(g^*_{1/2,\kappa} * \check{g}^*_{1/2,\kappa}) \geq 0$ for the specific family constructed here, which is a necessary but not sufficient condition for RH in the absence of a density argument for this family in S_{ad}.*

Computational note

All numerical claims are reproducible. Code is available at [2]. The Gaussian parameter $\varepsilon = 0.05$ is required for meaningful numerical experiments with the zero sum (for $\varepsilon = 0.5$ one has $e^{-\varepsilon^2\gamma_1^2} \approx 10^{-22}$, effectively suppressing all zero contributions).

Numerical comparison Z_{ren} vs $H^{\mathrm{ren}}_{\mathrm{local}}$. With $c_p^{\mathrm{ren}} = f_p^{1/2}$, $\varepsilon = 0.05$, and $N = 100$ zeros:

κ	Z_{ren}	$H^{\text{ren}}_{\text{local}}$	$Z - H$	$Z \geq H$
10	4.888	3.044	+1.844	✓
20	3.782	4.770	−0.988	×
29	5.520	3.588	+1.931	✓
50	5.693	2.881	+2.812	✓
100	5.311	1.562	+3.748	✓
200	5.102	2.355	+2.747	✓
500	5.156	1.101	+4.056	✓
1000	4.289	0.426	+3.863	✓

Remark. *The table illustrates the sawtooth behavior of* $H^{\text{ren}}_{\text{local}}$*: at each new prime* p*,* $H^{\text{ren}}_{\text{local}}$ *jumps upward by* $f_p > 0$*, then decreases continuously until the next prime. This creates temporary inversions* $Z_{\text{ren}} < H^{\text{ren}}_{\text{local}}$ *directly after a prime jump exactly as* $\pi(x) - \text{Li}(x)$ *fluctuates near small primes before stabilizing asymptotically.*

In our experiments, violations occur for $\kappa \leq 25$ *(in the range between the primes 23 and 29). For all* $\kappa \geq 26$*,* $Z_{\text{ren}} > H^{\text{ren}}_{\text{local}}$ *holds consistently, with the ratio* Z/H *growing from* 1.5 *at* $\kappa = 29$ *to over* 10 *at* $\kappa = 1000$*. This asymptotic stability is consistent with the spectral inequality* (6)*, though an analytic proof remains open.*

References

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